Lab-5

Static Code Analysis — Issue Documentation

Name: Chaithanya B **SRN:** PES1UG24CS123 **Section:** B

Issue table · 1 of 2

Issue	Type	Line(s)	Description	Fix approach
B201: Flask debug mode	Security High	80	Debug mode exposes an interactive debugger if reachable.	Set debug=False.
B506: Unsafe YAML loading	Security Medium	21	The YAML loader can construct arbitrary Python objects.	Replace yaml.load() with yaml.safe_load().
B612: Remote logging configuration	Security Medium	72	Remote logging configuration introduces an unnecessary code-execution risk.	Replace the listener with local logging.basicConfig().
B106: Hardcoded password	Security Low	66	A password is stored directly in the source code.	Read it from an environment variable and reject a missing value.
B311: Predictable lottery randomness	Security Low	56	Standard pseudo-random sampling is unsuitable when unpredictability matters.	Use SystemRandom from secrets to sample the student IDs.
W0102: Mutable default argument	Code quality	29	A default list is shared across calls and could cause bugs if modified.	Use None as the default and initialize a list inside the function.
C0123: Type checking	Code quality	30	Direct type comparison is unidiomatic.	Use isinstance() and explicitly reject boolean year values.
W0613: Unused gender argument	Logic / code quality	29–38	Room allocation ignores gender compatibility.	Check the room gender before allocating a bed.

Line numbers refer to the original, unmodified hostel_allocation.py.

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Issue	Type	Line(s)	Description	Fix approach
Room capacity check	Logic	34	Using > allows a full room to receive another occupant.	Use >= to skip rooms that have reached capacity.
Duplicate allocation	Logic	29–39	Repeated allocation can add the same student more than once.	Return the existing allocation before assigning another bed.
Incomplete vacation	Logic	41–45	Deleting an allocation does not remove the student from the room occupants.	Remove both the allocation and the occupant entry.
Incorrect seniority ordering	Logic	52–53	Ascending year order puts juniors before seniors.	Sort by year with reverse=True.
C0206: Dictionary iteration and vacant beds	Code quality / logic	58–63	The vacant-bed count excludes partially occupied rooms.	Iterate over room values and sum capacity minus occupancy.
C0103 / C0114 / C0116: Naming and documentation	Code quality	Various	Function names and documentation do not follow Python conventions.	Use snake_case names and add module and function docstrings.
W0611 / C0301: Unused import and long line	Code quality	2, 95	An unused import and an overly long expression reduce clarity.	Remove sys and split the expression into readable steps.
E302 / E305: Top-level spacing	Style	Various	Top-level definitions and the final statement have insufficient blank-line separation.	Add two blank lines between top-level definitions and after the final function.
E501: Long lines	Style	25, 66, 95	Some lines exceed Flake8's default 79-character limit.	Split dictionaries, calls, and expressions across shorter lines.

Logic-only findings were identified through code review and program output, not necessarily by the analysis tools.